

AFRYSTAR / ASSESSMENT OF DOUBLE ROMAN TILE BREAKAGE

(291121)

The breakage of the tile, as you have indicated to me in the wasap has not happened in the depalleting nor in the packing, because the breakage is produced by a different type of stresses or tensions that the concrete tile suffers, in the transit of people on the roof tiles, they are different.

Although there is some breakage of the tile in the corners, according to the photos provided, most of them are longitudinal breaks in the centre of the tile.

I am going to explain what I consider to be "the problem of these roof tiles" on a working hypothesis, which can be confirmed later with the data that can be taken from the roof tile.

As a reference I take the photo you sent me of the 2 roof tiles



During the manufacturing of the concrete tiles, there is a wear of the Small Slipper, Small Rollers, Big Slippers and Big Rollers, which gradually change the upper profile of the tile, causing an increase in the thickness and weight of the concrete tile.

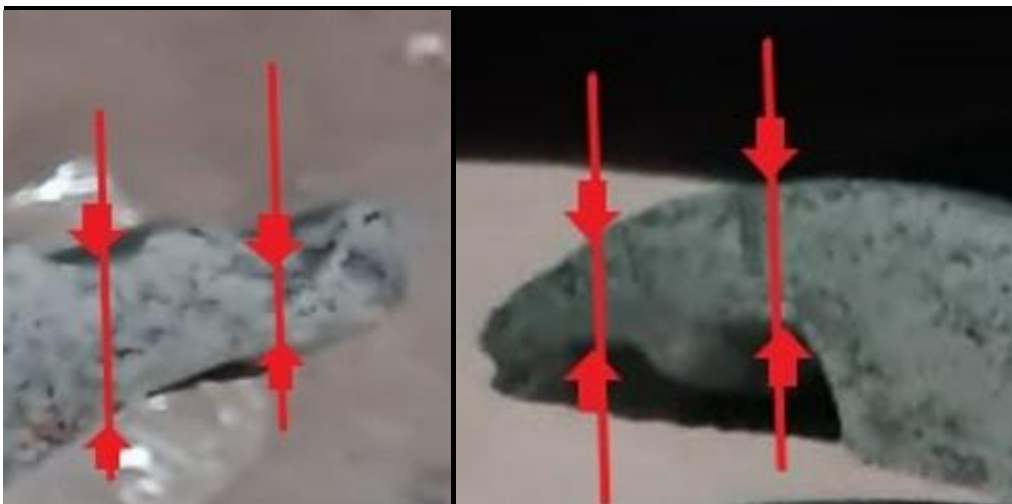


I believe that the measurements of the entire body of the tile must be very strictly controlled.



As is normal, there is first a rapid wear of the interlock with the Roller and the Small Slipper, and then in the body of the tile by the Big Slipper and Big Roller.

A) A) When the thicknesses of the interlock areas are greater than the permitted tolerance, we can have a problem of coupling between tiles.



Causing a bad coupling



Try to get a good coupling, and there should always be a certain amount of play in the interlock, such as when the Roller and Small Slipper are new.



The problem with this bad coupling between the tiles, is that they can lift the tile on one side of the tile (the tile is suspended in the air), not being able to support the tile on the "4" Front / Rear ribs and with the transit of people on the roof tiles, when stepping on the tiles, the corners or the tile break longitudinally.



In the case of the photos of the roof tiles, I consider that the tiles mounted on the roof are not supporting the "4" Front / Rear ribs of the tile.



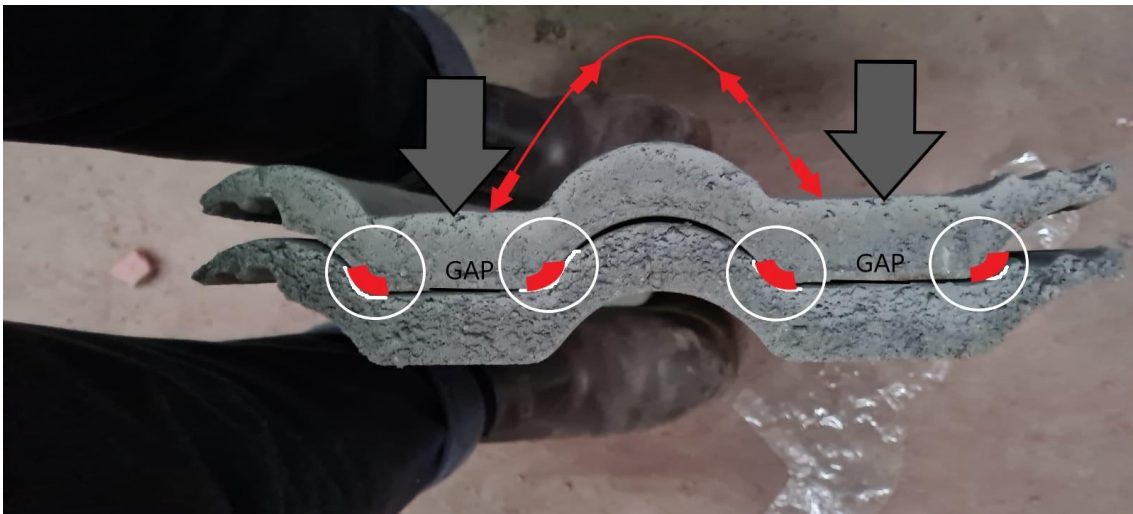
As there is an unsupported gap, the tile does not rest on the "4" Front / Rear ribs, and when the tile is stepped on, it breaks longitudinally.



- B) The upper part of the roof tile can also suffer from excessive wear in the Big Slipper, such as curved areas, these areas can cause bad support between the roof tiles and cause longitudinal breakage.



In these curved areas there must always be sufficient clearance, so that there is always support for "4" Front / Rear ribs of the tile and does not allow one tile to swing over the other.



Arganda del Rey 291121